

为什么采样信号频谱是周期性的？

Proof:

$$\begin{aligned}
 \frac{1}{2\pi} \int_{-0.5}^{0.5} X(f) \cdot e^{j2\pi fn} df &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_a(f) \cdot e^{j2\pi f(\frac{n}{F_s})} df \\
 &= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \int_{-\frac{1}{2}F_s + kF_s}^{\frac{1}{2}F_s + kF_s} X_a(f) \cdot e^{j2\pi f/F_s} df \\
 &= \frac{1}{2\pi} F_s \sum_{k=-\infty}^{\infty} \int_{-0.5+k}^{0.5+k} X_a\left(\frac{f}{F_s}\right) \cdot e^{j2\pi n \cdot (\frac{f}{F_s})} d\left(\frac{f}{F_s}\right) \\
 &= \frac{1}{2\pi} F_s \cdot \sum_{k=-\infty}^{\infty} \int_{-0.5}^{0.5} X_a\left(\frac{f}{F_s} - k\right) \cdot e^{j2\pi n(\frac{f}{F_s} - k)} d\left(\frac{f}{F_s}\right) \\
 &= \frac{1}{2\pi} F_s \cdot \int_{-0.5}^{0.5} \sum_{k=-\infty}^{\infty} X_a(f - k) \cdot e^{j2\pi n f} df
 \end{aligned}$$

$$\Rightarrow X(f) = F_s \cdot \sum_{k=-\infty}^{\infty} X_a(f - k)$$

—— 采样
..... 原始

时域离散，频域周期性。

时域周期性，频域离散